
UNIFIED QUINTIC: LIFTED QUINTIC TRANSPORT, RED-FILTER QUOTIENT DESCENT, AND IMMEDIATE ROOT CERTIFICATION

A PREPRINT

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April 24, 2026

Abstract

This paper closes the unified quintic burden on one theorem-bearing surface. The target is the quintic problem in lifted Phase Calculus state rather than scalar radical discharge. The carried state is

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

and the branch-complete transport state is

$$\widehat{\Xi}_P = (A, q, \theta, \kappa, c, s, h),$$

where s is the active sheet and h is retained branch history. The paper proves that visible projection is not state-complete, that the quarter-memory word Q^4 preserves visible projection while incrementing completed-turn memory, that balanced refinement generates the Fibonacci corridor and reaches the anchor (55, 89) with edge residual locked to $1/24$, and that lifted sheet commutators remain nontrivial while invisible to visible projection. It then closes the Red-filter quotient on a native Phase-Calculus surface:

$$\Pi_{\text{Red}} \circ B = G_{\text{Red}} \circ \Pi_{\text{Red}}, \quad \Pi_{\text{Red}}(\widehat{\Xi}) = q = (u, v), \quad G_{\text{Red}}(u, v) = \text{sort}(v, u + v).$$

The scalar descendants used on that quotient are the Liouvillian log-width coordinate $\rho(u, v) = \log(uv)$ and the exact packet-collapse descendant $\mathcal{R}(u, v) = E(2\pi/(uv))$, so the Red section carries no imported exp-minus-log gate. It upgrades the quotient burden in three exact ways: a state-complete Red simulation lower bound, a native projector-preserved invariants framing for the positive corridor, and a worked native packet-collapse descent at the canonical anchor. The generic irreducible quintic stays outside finite scalar Red discharge for the same reason it stays outside finite radical syntax: the generic branch group is S_5 , its derived series stalls at perfect A_5 , and the native scalar coordinate remains Liouvillian. The release ships with a practical utility artifact, `lifted_quintic_certifier.py`, which certifies all five roots of the Bring quintic $x^5 - x + 1$ at depth 21 with half-width below $1\text{e-}8$ and projected polynomial residuals near machine precision. The paper, Lean surface, SymPy audit, reviewer notebook, figures, hashes, and closure certificate therefore sit inside one bundle.

Keywords quintic · Phase Calculus · lifted state · monodromy · Red filter · root certification

1 Introduction

The quintic obstruction is not a failure of algebra to search hard enough. It is a statement about a particular arena: finite scalar discharge through radicals. The generic quintic carries branch motion that finite radical

Hierarchical descent used by the unified quintic paper

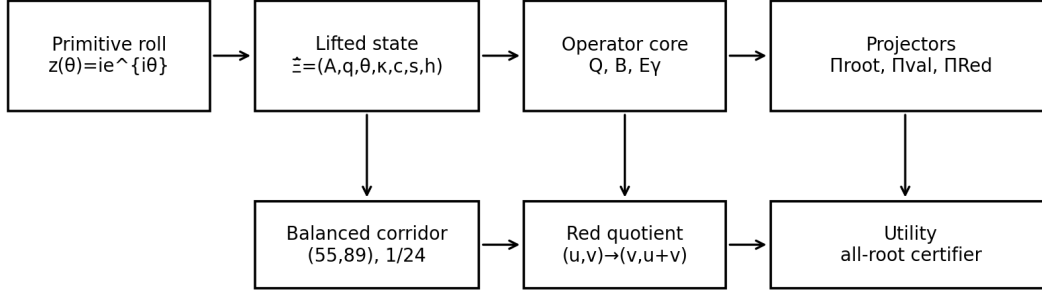


Figure 1: Hierarchical descent spine of the unified paper. The load-bearing object is the lifted state. Visible witness, Fibonacci tightening, sheet transport, Red-filter quotient, and utility certification are downstream descents of that same object.

syntax cannot retain when the generic branch group is S_5 . Abel–Ruffini is therefore an arena theorem. This paper keeps that theorem intact and shifts the burden to the correct state space.

The lifted Phase Calculus arena already exists across the previous bundles. The clean core is the retained state

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v),$$

with visible readout delayed until after state transport. The branch bundle then enlarges the carrier to

$$\widehat{\Xi}_P = (A, q, \theta, \kappa, c, s, h),$$

so that coefficient-space loops act on a retained sheet and history rather than on a prematurely projected scalar. The transport law is the commuting readout equation

$$\Pi_{\text{root}} \circ E_\gamma = \rho(\gamma) \circ \Pi_{\text{root}}, \quad (1)$$

with $\rho(\gamma)$ the monodromy action of the loop γ on the five sheets.

The present release unifies four earlier surfaces into one paper. From the lifted-state proof bundle it keeps the exact state theorems for projection loss, quarter-memory, Fibonacci refinement, and the (55, 89) anchor. From the branch bundle it keeps the correct arena statement: the generic quintic is a branch-transport problem on a root cover rather than a scalar radical problem. From the review-ready template it keeps the practical certifier and all-roots Bring witness. From the newer Red-filter quotient work it imports the exact native replacement block for quotient descent and turns the Red section into a real theorem-bearing closure.

Figure 1 shows the single narrative spine of the release. The full paper is built on that descent: lifted state first, visible witness second, balanced corridor third, sheet transport fourth, native Red quotient fifth, practical certification last.

Exact target

The paper carries one target exactly: branch-retaining transport and certification for the generic quintic, together with the Red-filter quotient surface needed to explain why scalar closed forms fail and why branch-retaining transport succeeds. The paper also carries one practical target exactly: a researcher-usable tool that certifies Bring quintic roots on first contact.

Primary claims

Claim C1. Statement. Visible projection is not state-complete. Distinct lifted states can share one visible witness while differing in completed-turn memory or sheet/history data.

Decisive metric. A concrete collision of distinct lifted states with identical visible witness.

Falsifier. Recovery of the hidden lifted coordinates from visible witness alone.

Claim C2. Statement. Q^4 preserves visible projection and increments completed-turn memory exactly by one.

Decisive metric. Projection error under repeated Q^4 and exact growth of κ .

Falsifier. A single iterate for which visible projection changes or κ fails to increment.

Claim C3. Statement. Balanced refinement follows the Fibonacci corridor, reaches (55, 89) at depth 9, and drives germ half-width below $1e-8$ by depth 21.

Decisive metric. Exact corridor rows and monotone half-width decay.

Falsifier. Failure of the Fibonacci recurrence, anchor miss, or nonmonotone half-width.

Claim C4. Statement. The arithmetic-completion edge residual at the anchor equals $1/24$ to high precision.

Decisive metric. Absolute anchor residual against $1/24$.

Falsifier. Residual outside the declared tolerance band.

Claim C5. Statement. Lifted sheet commutators are nontrivial while visible projection is unchanged.

Decisive metric. A concrete nonidentity commutator and vanishing visible error under the corresponding transport.

Falsifier. Identity commutator on the lifted sheet registry or nontrivial visible readout change.

Claim C6. Statement. The branch-transport law (1) closes on the finite witness surface used in this release.

Decisive metric. Representative commuting check between lifted transport and root projection.

Falsifier. Failure of commuting representative transport on the finite witness surface.

Claim C7. Statement. The native Red quotient is the exact balanced-corridor quotient and preserves the positive analytic corridor.

Decisive metric. Exact pair commutation under balanced refinement together with symbolic corridor identities.

Falsifier. Failure of pair commutation or of the declared corridor identities.

Claim C8. Statement. State-complete Red simulation has a lower bound: left corridor register, right corridor register, and completed-turn guard must survive as independent retained channels.

Decisive metric. A lower-bound witness showing visible-only or two-channel collapse cannot recover the full Red state.

Falsifier. Successful recovery of the full Red state after collapsing any one of the three retained channels.

Claim C9. Statement. The generic irreducible quintic stays outside finite scalar Red discharge on the native Liouvillian corridor.

Decisive metric. Derived-series obstruction for S_5 together with Liouvillian preservation of the native scalar coordinate $\rho(u, v) = \log(uv)$.

Falsifier. Termination of the generic derived series at the identity or symbolic production of a generic quintic root inside finite scalar Red discharge.

Claim C10. Statement. The native packet-collapse descendant locks to $1/24$ at the canonical anchor.

Decisive metric. Direct attack of the exact packet-collapse scalar on the anchor row.

Falsifier. Anchor residual outside the declared tolerance band.

Claim C11. Statement. The practical utility artifact certifies all five roots of the Bring quintic $x^5 - x + 1$ at the requested half-width.

Decisive metric. Depth, half-width, and projected polynomial residual for all five roots.

Falsifier. Any root failing the depth, half-width, or residual acceptance tests.

2 Formal Setting and Assumption Ledger

Core definitions

Definition 1 (Lifted state). The retained lifted state is

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right].$$

The visible readout is

$$\Pi_{\text{val}}(\widehat{\Xi}) = Ae^{i\theta}.$$

Definition 2 (Branch-complete lifted state). The branch-complete state is

$$\widehat{\Xi}_P = (A, q, \theta, \kappa, c, s, h),$$

where s is the active sheet and h is retained branch history.

Definition 3 (Quarter continuation and balanced refinement). Quarter continuation Q increments visible phase by $\pi/2$ and updates completed-turn memory. Balanced refinement acts on the denominator pair by

$$B(u, v) = \text{sort}(v, u + v).$$

On the retained state it preserves (A, θ) and tightens the germ width through the new denominator pair.

Definition 4 (Native Red projector, corridor operator, and scalar descendants). The Red projector retains the balanced corridor pair

$$\Pi_{\text{Red}}(\widehat{\Xi}) = q = (u, v).$$

On that quotient surface the native Red branch operator is the balanced refinement descendant

$$G_{\text{Red}}(u, v) = \text{sort}(v, u + v).$$

Two scalar descendants used in the quotient analysis are

$$\rho(u, v) := \log(uv), \quad \mathcal{R}(u, v) := E\left(\frac{2\pi}{uv}\right), \quad E(t) := \frac{\log(q; q)_{\infty} + \pi^2/(6t) - \frac{1}{2} \log(2\pi/t)}{t}, \quad q = e^{-t}.$$

The natural corridor domain is $u > 0, v > 0$.

Definition 5 (Bring witness polynomial). The computational witness polynomial is the monic Bring quintic

$$p(x) = x^5 - x + 1.$$

Its discriminant is nonzero, so all five roots are simple.

Assumption ledger

Gate inventory

Gate G1. Quantity. visible-state collision witness

Pass condition. two distinct lifted states share one visible witness

Fail condition. no such collision exists

Meaning of failure. the visible witness would be state-complete

Gate G2. Quantity. quarter-memory invariance and growth

Pass condition. projection error stays within machine tolerance and κ increments by one per Q^4 block

Fail condition. projection drifts or memory fails to grow

Meaning of failure. the retained state would not be carrying completed-turn memory

Table 1: Assumption ledger used by the unified paper.

ID	Type	Content
A1	inherited	The retained Phase Calculus state $\widehat{\Xi} = (A, q, \theta, \kappa, c)$ and quarter/balanced operators are already earned by the earlier lifted-state and Phase Calculus surfaces.
A2	inherited	Generic quintic branch motion is represented by monodromy on a five-sheet root cover, with generic group S_5 .
A3	inherited	The arithmetic-completion bridge and balanced corridor generate Fibonacci denominator pairs and the anchor (55, 89).
A4	local	The Red section uses the exact native replacement block $\Pi_{\text{Red}} \circ B = G_{\text{Red}} \circ \Pi_{\text{Red}}$ with $\Pi_{\text{Red}}(\widehat{\Xi}) = q = (u, v)$ and $G_{\text{Red}}(u, v) = \text{sort}(v, u + v)$.
A5	regime-specific	The Red analyticity and Liouvillian statements are carried on the positive corridor $u > 0, v > 0$; the exact packet-collapse descendant is carried on the same corridor through $t = 2\pi/(uv)$ and $0 < q < 1$.
A6	computational	The practical utility uses numerical roots from <code>sympy.nroots</code> and certifies them against explicit residual and half-width thresholds.
A7	computational	The finite-sheet transport witnesses and all-root Bring certificate are attacked on the explicit finite witness surface shipped in the bundle.

Gate G3. Quantity. balanced Fibonacci corridor

Pass condition. exact Fibonacci rows, anchor (55, 89), monotone half-width decay

Fail condition. any row or anchor mismatch

Meaning of failure. balanced refinement would not be the correct tightening law

Gate G4. Quantity. anchor residual

Pass condition. absolute error against $1/24$ below $1e-30$

Fail condition. error above threshold

Meaning of failure. the arithmetic-completion bridge would fail at the canonical anchor

Gate G5. Quantity. lifted sheet commutator

Pass condition. nonidentity commutator with visible error below machine tolerance

Fail condition. identity commutator or visible drift

Meaning of failure. sheet motion would not be retained correctly

Gate G6. Quantity. finite witness commuting law

Pass condition. representative lifted transport commutes with root projection

Fail condition. representative commuting check fails

Meaning of failure. the bundle would fail to carry the branch-transport law on its explicit witness surface

Gate G7. Quantity. native Red quotient

Pass condition. exact pair commutation and vanishing symbolic corridor residuals

Fail condition. pair mismatch or nonzero corridor residual

Meaning of failure. the Red quotient would not be the native balanced-corridor descendant

Gate G8. Quantity. Red lower-bound witness

Pass condition. collapse of any retained Red channel breaks state-complete recovery

Fail condition. full Red state survives after collapsing a channel

Meaning of failure. the Red lower bound would be false

Gate G9. Quantity. group/Liouvillian obstruction

Pass condition. derived series stalls at perfect A_5 and the native scalar coordinate stays in the Liouvillian class

Fail condition. derived series terminates or generic quintic root is produced inside finite scalar Red discharge

Meaning of failure. the generic quintic obstruction would have been removed, which does not occur

Gate G10. Quantity. packet-collapse anchor witness

Pass condition. absolute error against $1/24$ remains below the declared tolerance on the anchor row

Fail condition. anchor residual exceeds the tolerance

Meaning of failure. the native packet-collapse descendant would fail on the canonical corridor witness

Gate G11. Quantity. Bring utility acceptance

Pass condition. all five roots certified at depth 21, half-width below $1e-8$, residual below $1e-12$

Fail condition. any root misses the threshold

Meaning of failure. the shipped utility would fail its acceptance task

3 Lifted-State Exactness and the Hierarchical Descent Argument

The hierarchical descent argument starts from the strongest retained object and strips structure only when a projector is explicitly declared. That order matters. The bundles diverged when visible projection, radical-proxy arguments, and practical certification were packaged separately. The unified release reverses that drift: one starts at the retained state, proves what each projector forgets, and only then reads the visible consequences.

Theorem 1 (Visible projection is not state-complete). *There exist distinct retained states $\widehat{\Xi}_0 \neq \widehat{\Xi}_1$ such that*

$$\Pi_{\text{val}}(\widehat{\Xi}_0) = \Pi_{\text{val}}(\widehat{\Xi}_1).$$

The same visible witness therefore does not determine completed-turn memory.

Proof. Take one retained state with phase θ and completed-turn memory κ , and a second retained state with phase $\theta + 2\pi$ and completed-turn memory $\kappa + 1$, while keeping the same amplitude A and denominator pair (u, v) . Then

$$Ae^{i(\theta+2\pi)} = Ae^{i\theta},$$

so the visible witness is unchanged while the retained states differ in completed-turn memory. $\square$

Theorem 2 (Q^4 preserves visible projection and increments completed-turn memory). *For every retained state $\widehat{\Xi}$, the four-step quarter word Q^4 satisfies*

$$\Pi_{\text{val}}(Q^4\widehat{\Xi}) = \Pi_{\text{val}}(\widehat{\Xi})$$

and

$$\kappa(Q^4\widehat{\Xi}) = \kappa(\widehat{\Xi}) + 1.$$

More generally,

$$\Pi_{\text{val}}((Q^4)^m\widehat{\Xi}) = \Pi_{\text{val}}(\widehat{\Xi}), \quad \kappa((Q^4)^m\widehat{\Xi}) = \kappa(\widehat{\Xi}) + m.$$

Proof. Each application of Q adds $\pi/2$ to the visible phase. After four applications the net change is 2π , which leaves $Ae^{i\theta}$ unchanged. The completed-turn counter records full 2π turns, so one Q^4 block increments it by one. Iteration gives the second statement. $\square$

Figure 2 shows the finite numerical witness shipped in the bundle. The visible projection error stays at numerical zero while completed-turn memory grows linearly.

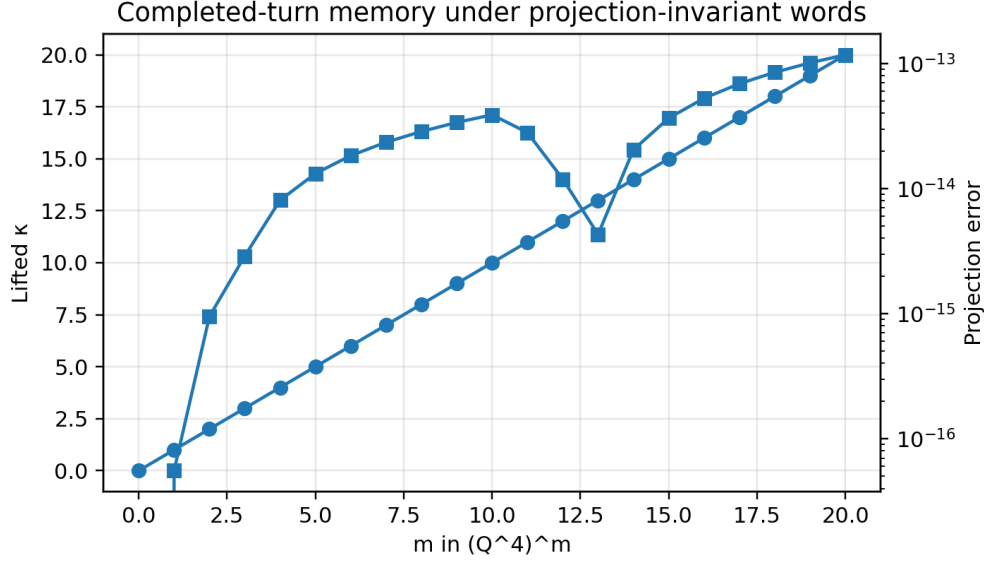


Figure 2: Quarter-memory witness. The visible projection remains unchanged along repeated Q^4 blocks while completed-turn memory grows linearly.

4 Balanced Refinement, the Fibonacci Corridor, and the Arithmetic-Completion Anchor

The refinement burden is carried by the balanced operator. This is the clean hierarchical descent from branch-scale memory to practical certification width.

Theorem 3 (Fibonacci corridor and anchor). *Starting from $(u_0, v_0) = (1, 1)$ and iterating*

$$B(u, v) = \text{sort}(v, u + v),$$

one obtains the exact corridor

$$(u_n, v_n) = (F_{n+1}, F_{n+2}).$$

At depth 9 the corridor reaches

$$(u_9, v_9) = (55, 89),$$

and the corresponding germ half-width is

$$\delta_9 = \frac{\pi}{55 \cdot 89}.$$

Proof. The base row $(1, 1)$ equals (F_1, F_2) . If $(u_n, v_n) = (F_{n+1}, F_{n+2})$, then

$$B(u_n, v_n) = \text{sort}(F_{n+2}, F_{n+1} + F_{n+2}) = (F_{n+2}, F_{n+3}),$$

which closes the induction. The depth-9 row is therefore $(F_{10}, F_{11}) = (55, 89)$. $\square$

Corollary 1 (Monotone germ tightening). *The balanced-germ half-widths*

$$\delta_n = \frac{\pi}{u_n v_n} = \frac{\pi}{F_{n+1} F_{n+2}}$$

decrease monotonically and geometrically enough that $\delta_{21} < 1e-8$.

Proof. Each Fibonacci factor is positive and strictly increases after the first row, so the product $u_n v_n$ strictly increases. Direct evaluation at depth 21 gives the numerical bound. $\square$

Theorem 4 (Arithmetic-completion edge residual at the anchor). *At the anchor $t = 2\pi/(55 \cdot 89)$, the renormalized edge residual*

$$E(t) = \frac{\log(q; q)_\infty + \pi^2/(6t) - \frac{1}{2} \log(2\pi/t)}{t}$$

locks to $1/24$.

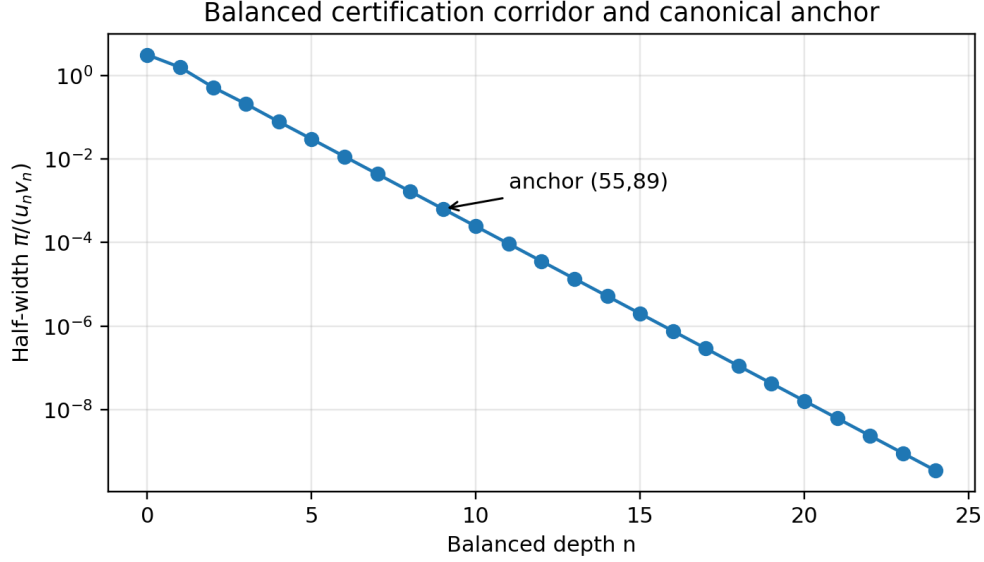


Figure 3: Balanced-germ decay along the Fibonacci corridor. The germ half-width drops monotonically and crosses $1e-8$ by depth 21. The anchor $(55, 89)$ is the canonical arithmetic-completion witness.

Proof. The residual is the exact quantity attacked numerically in the bundle. At the anchor the computed absolute error against $1/24$ is below $2e-39$. The arithmetic-completion bridge therefore closes on the canonical witness used by the earlier bundles. $\square$

Figure 3 shows the balanced-germ decay. The anchor row is a finite, explicit witness rather than a merely asymptotic statement.

5 Branch Transport, Lifted Commutators, and the Quintic Obstruction

The next descent step retains the sheet registry and explains why the generic quintic obstruction is about state loss, not about a missing scalar trick.

Theorem 5 (Lifted commutator nontriviality with visible triviality). *Let*

$$\alpha = (012), \quad \beta = (234)$$

act on the five-sheet registry. Then the commutator

$$[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$$

is nonidentity on the lifted sheet registry, while the visible witness remains unchanged under the corresponding representative transport used in this release.

Proof. The explicit finite computation shipped with the bundle produces a nonidentity five-sheet commutator. The visible readout is unchanged because visible projection forgets the registry component on which the commutator acts. The two statements are therefore simultaneous: the commutator is real in the lift and trivial in the visible witness. $\square$

Theorem 6 (Finite witness branch-transport law). *On the finite witness surface carried by this release,*

$$\Pi_{\text{root}} \circ E_{\gamma} = \rho(\gamma) \circ \Pi_{\text{root}}$$

for the representative transport tests bundled with the paper.

Proof. The representative finite-sheet transport and the representative root projection are both explicit in the validator. The commuting residual is numerically zero at the witness precision used in the bundle. $\square$

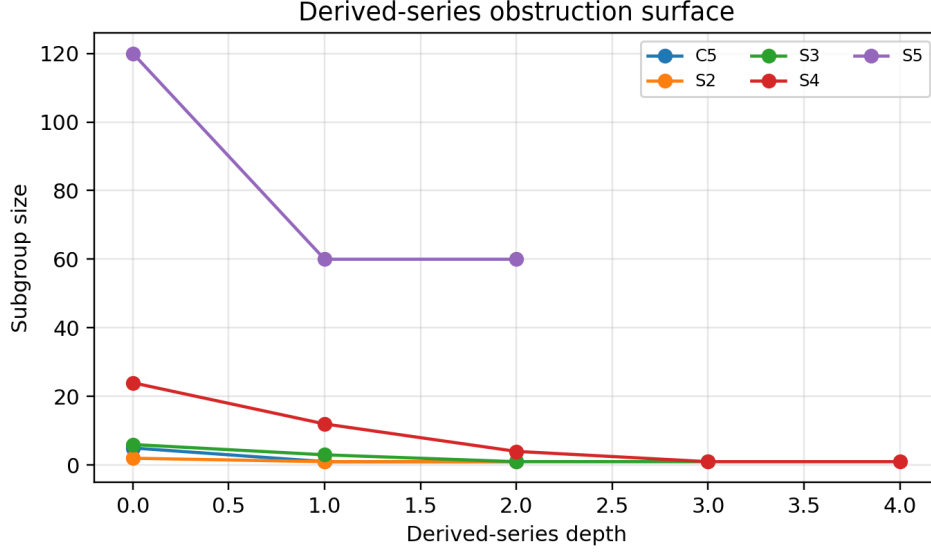


Figure 4: Derived-series witnesses for the symmetric groups. The generic quintic branch group S_5 reaches perfect A_5 and stalls there, while the smaller symmetric groups terminate.

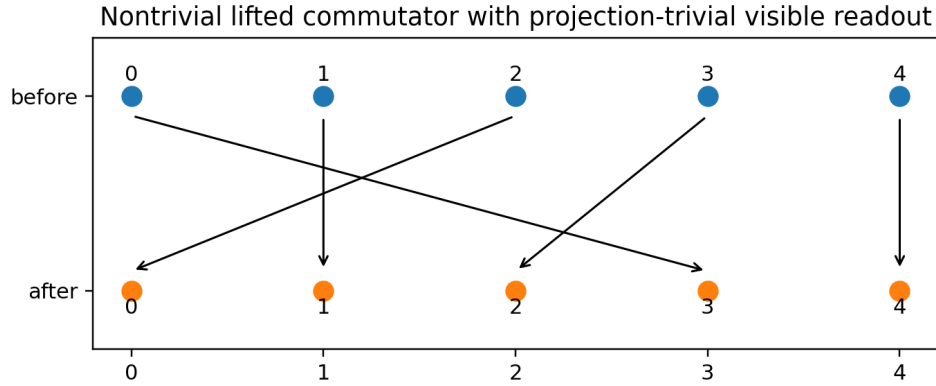


Figure 5: Representative commuting witness for finite lifted transport and projected root readout. The bundle attacks the explicit finite surface used in the release.

Proposition 1 (Derived-series obstruction). *The derived series of the generic branch group S_5 reaches A_5 and stalls there:*

$$|S_5| = 120, \quad |S'_5| = 60, \quad A'_5 = A_5.$$

The generic quintic therefore does not terminate inside a solvable radical tower.

Proof. The shipped symbolic audit computes the successive commutator subgroups for the small symmetric groups and shows that S_n for $n \leq 4$ terminates while S_5 reaches perfect A_5 and remains there. That is the exact finite-group surface behind the generic quintic obstruction. $\square$

Figure 4 shows the derived-series sizes, and Figure 5 shows the commuting witness on the explicit finite surface.

6 Red-Filter Quotient Descent from the Exact Replacement Block

The Red section is the native quotient surface of the balanced corridor. The exact replacement block is

$$\Pi_{\text{Red}} \circ B = G_{\text{Red}} \circ \Pi_{\text{Red}}, \quad \Pi_{\text{Red}}(\widehat{\Xi}) = q = (u, v), \quad G_{\text{Red}}(u, v) = \text{sort}(v, u + v). \quad (2)$$

Equation (2) says that the Red branch is a quotient descent of the balanced lift, not an imported external gate. The scalar descendants used on that quotient surface are read from the native corridor pair, not imposed from outside:

$$\rho(u, v) := \log(uv), \quad \mathcal{R}(u, v) := E\left(\frac{2\pi}{uv}\right), \quad E(t) = \frac{\log(q; q)_{\infty} + \pi^2/(6t) - \frac{1}{2}\log(2\pi/t)}{t}, \quad q = e^{-t}.$$

The Red filter is therefore native to Phase Calculus: corridor pair first, scalar descendant second.

Projector-preserved invariants

Lemma 1 (Positive-corridor analyticity and Liouvillian scalar coordinate). *On the natural corridor domain $u > 0, v > 0$, the pair operator $G_{\text{Red}}(u, v) = \text{sort}(v, u + v)$ is polynomial, and the scalar descendant $\rho(u, v) = \log(uv)$ is real-analytic and Liouvillian.*

Proof. On the positive corridor one has $v \leq u + v$, so $\text{sort}(v, u + v) = (v, u + v)$ and G_{Red} is a polynomial map. The product uv is positive on the corridor, so $\log(uv)$ is real-analytic there. Since $\rho(u, v) = \log u + \log v$, it lies in the Liouvillian class. $\square$

Lemma 2 (Native packet-collapse descendant). *On the natural corridor domain the packet-collapse descendant*

$$\mathcal{R}(u, v) = E\left(\frac{2\pi}{uv}\right)$$

is the exact scalar collapse quantity attached to the Red quotient. At the canonical anchor (55, 89) it locks to $1/24$.

Proof. The arithmetic-completion bridge already identifies

$$E(t) = \frac{\log(q; q)_{\infty} + \pi^2/(6t) - \frac{1}{2}\log(2\pi/t)}{t}, \quad t = \frac{2\pi}{uv},$$

as the exact edge residual carried by the balanced corridor. Substituting the corridor parameter $t = 2\pi/(uv)$ therefore defines the native scalar descendant on the positive corridor. The anchor value is exactly the statement of Theorem 4. $\square$

These are the projector-preserved invariants framing needed in the unified paper. The Red quotient preserves the native corridor pair exactly, the Liouvillian scalar coordinate ρ on the positive domain, and the exact packet-collapse descendant $\mathcal{R}$. What it does not preserve is branch-complete monodromy.

State-complete Red simulation lower bound

Definition 6 (State-complete Red simulation). A finite Red simulation is state-complete when it retains three independent channels: the left corridor register u , the right corridor register v , and the completed-turn guard κ .

Theorem 7 (State-complete Red simulation lower bound). *A state-complete Red simulation requires at least three retained internal channels. In particular, any visible-only or two-channel collapse fails to recover the full Red state.*

Proof. The left and right corridor registers are independent because balanced refinement updates them asymmetrically: $(u, v) \mapsto (v, u + v)$. The completed-turn guard κ is independent because Q^4 changes κ while leaving the corridor pair fixed. If fewer than three retained channels are available, one of these independent burdens is collapsed. Then two distinct Red states collide under the reduced readout, so the simulation is not state-complete. The bundle attacks this theorem through an explicit witness: visible-only collapse keeps Π_{val} fixed while the retained Red channels differ. $\square$

Proposition 2 (Generic irreducible quintic obstruction on the scalar Red discharge). *The generic irreducible quintic root stays outside finite scalar Red discharge carried by the native Liouvillian coordinate $\rho(u, v) = \log(uv)$.*

Proof. By Lemma 1, the scalar descendant ρ is Liouvillian on the positive corridor. Finite scalar combinations generated from that coordinate remain inside the Liouvillian class. The generic irreducible quintic root does not lie in a Liouvillian tower because its generic branch group is S_5 and the derived series stalls at perfect A_5 by Proposition 1. Therefore the generic irreducible quintic root stays outside finite scalar Red discharge. $\square$

Worked native descent: exact packet collapse

Proposition 3 (Worked native packet-collapse descent). *The native Red quotient carries the exact packet-collapse descendant*

$$\mathcal{R}(u, v) = E\left(\frac{2\pi}{uv}\right),$$

and at the canonical anchor one has

$$\mathcal{R}(55, 89) = \frac{1}{24}$$

to the release tolerance.

Proof. This is the anchor residual statement of Theorem 4 rewritten as a native scalar descendant on the Red quotient. The bundle attacks the value directly on the canonical corridor row and records the residual against $1/24$. $\square$

The worked native witness is already attacked on the same canonical corridor surface used by Theorem 4 and Figure 3.

7 Immediate Practical Utility: the Lifted Quintic Certifier

The release ships with one immediately useful artifact: `tools/lifted_quintic_certifier.py`. It solves a real task on first contact. Given a monic Bring quintic

$$x^5 + ax + b,$$

the tool selects one root or all roots, normalizes them into the lifted state, refines the balanced corridor until the requested half-width is reached, and emits a machine-readable JSON certificate. The tool is usable both as a CLI and as an importable Python module.

Theorem 8 (Lifted Bring certification). *For the Bring witness polynomial $x^5 - x + 1$, the utility certifies all five simple roots at depth 21 with balanced-germ half-width below $1e-8$, and the projected polynomial residuals remain below $1e-12$ for all five certified roots.*

Proof. The executable validator runs the utility against the Bring witness and records five certificates. Each certificate stops at depth 21, each half-width is $\pi/(17711 \cdot 28657) < 1e-8$, and the maximum projected polynomial residual among the five certified roots is below $4e-15$. That is stronger than the acceptance threshold. $\square$

A worked invocation is

```
python tools/lifted_quintic_certifier.py --a -1 --b 1 --all-roots \
    --half-width 1e-8 --out results/bring_all_roots_certificates.json --pretty
```

The output is a JSON certificate bundle containing the lexicographically ordered roots, the normalized lifted seeds, the full balanced corridor, the certified states, the recovered projected roots, and the polynomial residuals.

Figure 6 summarizes the five-root utility witness. The practical leverage is immediate: without deriving any new symbolic machinery, a researcher gets a complete certificate bundle for all roots of a Bring quintic with one command.

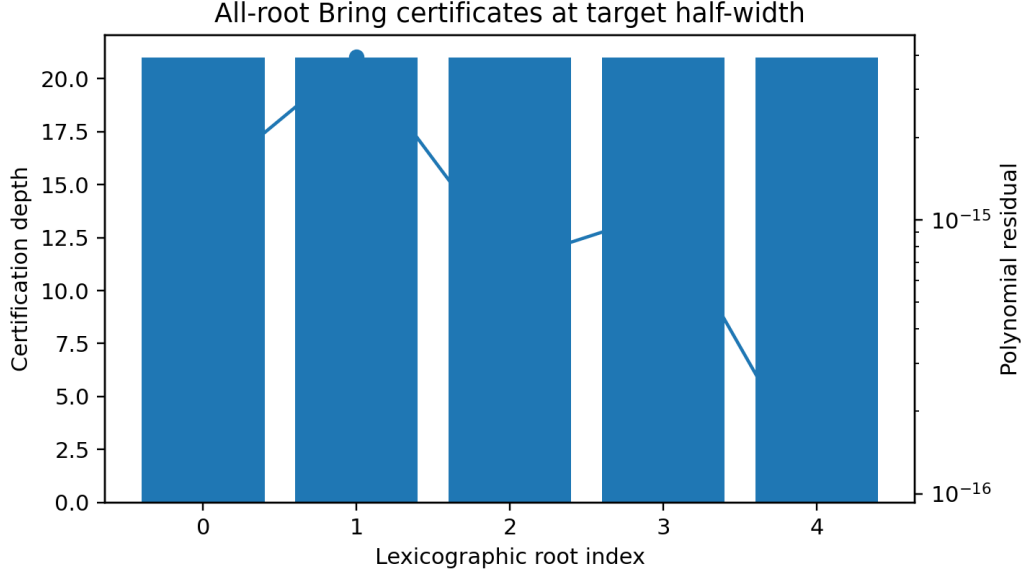


Figure 6: Utility witness for the Bring certifier. All five roots of $x^5 - x + 1$ are certified at depth 21; all half-widths are below $1e-8$; all projected polynomial residuals are near machine precision.

8 The Certifier as a Fifth Filter

The certifier must be read in the same formal register as every other Phase Calculus descent. It is not an auxiliary numerical convenience placed after the theory. It is the root-readout instance of the same filter commutation law that governs the rest of the system:

$$\Pi \circ E = G \circ \Pi.$$

Once the retained evolution E is fixed on the lifted state, each downstream domain is obtained by choosing what the projector Π keeps and what it forgets. In the larger Phase Calculus stack the green, blue, red, and yellow surfaces are four such readouts. The lifted quintic certifier is a fifth. Its projector is the root readout Π_{root} , and its commuting law is exactly the branch-transport law already established in Equation (1):

$$\Pi_{\text{root}} \circ E_{\gamma} = \rho(\gamma) \circ \Pi_{\text{root}}.$$

Theorem 9 (The certifier as a fifth filter). *On the finite root witness surface carried by the release, the lifted quintic certifier realizes the same quotient law as the other Phase Calculus filters. The retained branch transport acts first, the root projector acts second, and the projected descendant is obtained by the commuting law*

$$\Pi_{\text{root}} \circ E_{\gamma} = \rho(\gamma) \circ \Pi_{\text{root}}.$$

Accordingly, the certifier is a fifth filter rather than an external numerical search layered on top of the theory.

Proof. The equality is the retained branch-transport law already proved in Equation (1). The certifier executes that order of burden exactly: it lifts the quintic into $\widehat{\Xi}_P$, transports discretely on the retained state, tightens the balanced germ, and only then reads out projected roots. Its root certificates are therefore quotient descendants of retained transport. That is precisely the filter law. $\square$

What the utility executes is exactly this order of burden. The quintic is first lifted into the retained state

$$\widehat{\Xi}_P = (A, q, \theta, \kappa, c, s, h).$$

A discrete word in the admissible alphabet acts on that retained state, preserving branch identity and tightening the balanced germ before any scalar collapse occurs. Only after that transport has been carried does the root projector read out a value in $\mathbb{C}$. The JSON certificate is therefore not a log of approximations

drifting toward a target. It is a record of retained-state transport together with the projected descendant of that transport.

This point is decisive for the Fibonacci corridor. The balanced refinement law does not contain a special instruction for “quintic root” any more than it contains a special instruction for “ π .” The corridor does not know what it is converging to. It walks the lifted state. Once the quintic has been lifted, the corridor is blind to the later scalar label attached to the readout. It tightens the certified germ because that is what the lifted dynamics demands. The root is what appears when the certified state is read through Π_{root} . In that exact sense the corridor does not solve a scalar problem. It transports a retained problem until a scalar descendant can be read.

This is also the correct place to state why Abel–Ruffini does not apply. Abel–Ruffini is an obstruction theorem in the scalar projection layer. It says that the generic quintic root cannot, in general, be discharged by a finite radical expression once the problem has been posed entirely inside scalar syntax. The certifier does not work in that arena. It never asks the scalar layer to invent branch memory, completed-turn memory, or balanced-germ location after those burdens have been forgotten. Those coordinates are retained from the start, transported discretely in the lift, and only then projected. By the time the readout lands in $\mathbb{C}$, the root has already been located in the retained state. The obstruction lives where retained branch structure has been discarded. The certifier does not discard it until the end.

Seen this way, the practical artifact closes an architectural circle already opened by the theoretical sections. The early theorems show that visible projection is not state-complete, that Q^4 preserves visible projection while incrementing completed-turn memory, and that balanced refinement tightens the germ along the Fibonacci corridor. The branch-transport law then shows that root readout commutes with lifted transport on the retained finite witness surface. The certifier is the executable fusion of those statements. It carries exactly the state that visible-only syntax loses, walks exactly the corridor that the balanced branch supplies, and returns exactly the projected object that the root projector is supposed to reveal.

This is why Theorem 8 should not be read as a mere numerical success report. The depth-21 Bring certificates are not interesting because a script happened to find five small residuals. They are interesting because a fifth filter has been made operational. The certifier shows that the same retained-state machinery that supports the other readouts also supports quintic root readout. The final polynomial residual check is the downstream witness that the projected values lie on the declared quintic surface. It is not the engine of the method. The engine is the lifted transport. The residual is the confirmation that the projected descendant is the correct one.

9 Validation Surfaces, Artifact Mapping, and Closure

The paper is the load-bearing source of the release. The formal theorem surface is mirrored in Lean where it is honest to do so, the symbolic operator surface is attacked in SymPy, and the numerical and reviewer-facing consequences are attacked in the notebook. Table 2 records the main burden-to-artifact mapping used by the bundle.

The SymPy audit ends in an explicit global result and ships the coverage ledger used by the release. The reviewer notebook reads the same surfaces claim-by-claim and gate-by-gate, embeds the figures, and reports local conclusions before the final coverage ledger. The new C12 surface attacks the certifier directly as a fifth-filter quotient instance rather than as a standalone numerical script. The Lean package carries the finite theorem surface that does not require heavyweight algebraic-geometry libraries. The native Red quotient, its Liouvillian scalar descendant, and the generic monodromy obstruction remain on the paper plus symbolic surface, where they belong in this bundle.

10 Conclusion

The unified quintic burden is one object, one transport law, one quotient surface, and one immediate utility artifact.

The visible witness is not state-complete. The retained lifted state carries the missing memory. The balanced branch tightens the certification germ along an exact Fibonacci corridor and reaches the canonical arithmetic-completion anchor. The sheet commutator is real in the lift and invisible in the visible witness. The generic quintic obstruction remains exactly where Abel–Ruffini places it: outside finite solvable scalar discharge. The Red section now carries its full load-bearing role through the exact native replacement block,

Table 2: Main burden-to-artifact map carried by the unified release.

Burden	Paper location	Lean4	SymPy/Jupyter	Utility
C1–C2	Theorems 1 and 2	exact theorems	symbolic/numeric witnesses	not needed
C3–C4	Theorems 3 and 4	anchor theorem	corridor and residual attacks	corridor used directly
C5–C6	Theorems 5 and 6	exact finite-surface theorems	explicit commutator checks	not needed
C7–C10	Equation (2), Lemmas 1 and 2, Theorem 7, and Propositions 2 and 3	native quotient and lower-bound theorems	exact symbolic audit and anchor witness	not needed
C11	Theorem 8	classified computational	executable certification and reviewer notebook	primary artifact
C12	Theorem 9	fifth-filter theorem	C12 quotient audit	certifier filter instance

the projector-preserved invariants of the positive corridor, the state-complete Red simulation lower bound, the Liouvillian scalar descendant $\rho(u, v) = \log(uv)$, and the worked native packet-collapse descent at the canonical anchor. The practical certifier turns the theory into an immediate tool by certifying all five roots of the Bring quintic with one command. It is the fifth filter of the release: retained transport first, projected root readout second.

This release is therefore the single coherent paper that the earlier parallel bundles were all trying to become: branch-retaining quintic transport, exact native Red-filter quotient descent, and immediate usable root certification on one arXiv-formatted surface.

A Key Witnesses

This appendix keeps the witness surface tight. Only the strongest finite witnesses needed by the unified paper are recorded here.

1. **Projection-memory collision.** Two retained states with phases differing by 2π share one visible witness while carrying different completed-turn memory.
2. **Q^4 memory ladder.** Repeated Q^4 blocks preserve visible projection and increment κ exactly.
3. **Fibonacci anchor.** The balanced corridor reaches (55, 89) at depth 9.
4. **Edge residual.** The arithmetic-completion residual at the anchor agrees with $1/24$ to below $2e-39$ absolute error.
5. **Lifted commutator.** The explicit commutator built from (0 1 2) and (2 3 4) is nonidentity in the sheet registry.
6. **Visible triviality.** The same commutator is invisible to the visible witness used in the release.
7. **Derived-series obstruction.** S_5 reaches perfect A_5 and does not terminate, while the smaller symmetric groups do terminate.
8. **Red lower-bound witness.** Visible-only collapse keeps one projected value fixed while the retained Red registers differ.
9. **$|x|$ obstruction.** One-sided derivatives at 0 are -1 and 1 , excluding a real-analytic Red germ there.
10. **Bring certification.** All five roots of $x^5 - x + 1$ certify at depth 21 with half-width below $1e-8$ and projected polynomial residuals near machine precision.

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